

- c. Explain the detection of amplitude modulated signal using diode detector.

SECTION – II

Q - 4 Define or describe following terms. [07]

- a. White noise
- b. Noise factor
- c. Sensitivity of receiver
- d. SNR
- e. Modulation
- f. Bandwidth
- g. Product detector

Q – 5.a Prove Fourier transform of Autocorrelation function is ESD of energy signal. [05]

OR

Q – 5.a Prove Fourier transform of Autocorrelation function is PSD of power signal. [05]

Q – 5.b Find the Fourier Transform of function $e^{at}u(t)$. [04]

Q – 5.c Derive the Friis's formula for the noise factor when the amplifiers are connected in cascade. [05]

OR

Q – 5.c What is noise? Explain the different types of noise. [05]

Q – 6.a The equivalent noise resistance for an amplifier is 200Ω , and the equivalent shot noise current is $4\mu A$. The amplifier is fed from a 150Ω , $10\mu V$ rms sine signal source. Calculate the individual noise voltages at the input and input SNR in decibels. The bandwidth is 20Mhz [05]

OR

Q – 6.a Two resistors of $10K\Omega$ and $60K\Omega$ are at the room temperature. For a bandwidth of 150 KHz, calculate the thermal noise voltage generated by (1) each resistor (2) two resistors in series (2) two resistors in parallel. [05]

Q – 6.b Write short note on Modulation Index For Sinusoidal AM. [04]

Q – 6.c Draw the block diagram of Superhetrodyne Receiver and explain function of each block in detail. [05]

OR

Q – 6.c Compare Amplitude modulation and Frequency modulation. [05]
